SCRIPT

Reducing Corruption Through Blockchain-Based Transparency Systems

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SLIDE 1 – TITLE (Jonathan & William)

Good morning,

Today, we would like to present our research entitled *Reducing Corruption Through Blockchain-Based Transparency Systems*.

This research was conducted by Jonathan Theja, William Irawan, rom the School of Computer Science, BINUS University.

SLIDE 2 – BACKGROUND & PROBLEM (William)

Corruption remains a major challenge in Indonesian governance.

Despite having comprehensive anti-corruption laws, Indonesia still faces weak transparency and limited accountability in public administration.

One of the main causes is the heavy reliance on centralized information systems, where data can be manipulated internally without leaving clear audit trails.

This situation reduces public trust and creates opportunities for corruption. Therefore, a more transparent and tamper-proof system is urgently needed.

SLIDE 3 – RESEARCH OBJECTIVES (William)

Based on these problems, this research has several objectives.

First, we aim to analyze the weaknesses of centralized data management systems commonly used in e-government.

Second, we propose a blockchain-based transparency system as an alternative solution.

Finally, we compare centralized and blockchain-based systems to evaluate their performance in terms of data integrity, transparency, and resistance to manipulation.

SLIDE 4 – LITERATURE REVIEW OVERVIEW (William)

From previous studies, corruption is defined as the abuse of authority for personal or institutional gain, which negatively impacts governance and economic development.

Transparency is widely recognized as a key factor in preventing corruption, as it enables accountability and public oversight.

Blockchain technology has been highlighted in many studies as a promising tool due to its decentralized nature, immutability, and cryptographic security.

Several researchers have also demonstrated blockchain's potential in enhancing transparency within public sector governance.

SLIDE 5 – BLOCKCHAIN CONCEPT & ARCHITECTURE (William)

Blockchain is a decentralized digital ledger where transactions are recorded permanently and cannot be altered once validated.

Unlike centralized databases, blockchain distributes data across multiple nodes, eliminating a single point of control.

This ensures transparency, traceability, and data integrity.

In this research, blockchain serves as the core technology to prevent unauthorized data manipulation and strengthen accountability in public records.

SLIDE 6 – RESEARCH METHODOLOGY (William)

This study uses an experimental research design.

We developed two prototype systems for comparison.

System A represents a traditional centralized architecture, while System B is a blockchain-based decentralized system.

Both systems were tested under controlled conditions, and their performance was evaluated based on transparency, integrity, and auditability.

SLIDE 7 – SYSTEM A: CENTRALIZED SYSTEM (Jonathan)

Now, I will explain the prototype implementation, starting with System A.

System A represents a traditional centralized system developed using Python Flask and RESTful APIs.

All data is stored in a centralized in-memory database.

This system allows users to insert, retrieve, and modify data freely, which simulates how typical e-government systems operate today.

SLIDE 8 – SYSTEM B: BLOCKCHAIN SYSTEM (Jonathan)

In contrast, System B is a blockchain-based system implemented using Solidity smart contracts and deployed on the Ethereum Sepolia Testnet.

The system is integrated with MetaMask wallet for user authentication.

Once data is submitted and confirmed, it cannot be modified or deleted.

This design ensures immutability, transparency, and decentralized control.

SLIDE 9 – RESULTS COMPARISON (Jonathan)

"When we compare both systems, the differences are very clear.

In System A, data manipulation is possible and changes leave no audit trail.

Meanwhile, in System B, all transactions are immutable and publicly verifiable through the blockchain.

This demonstrates that blockchain provides a much stronger level of transparency and integrity compared to centralized systems."

SLIDE 10 – SYSTEM A RESULTS (Jonathan)

In System A, data manipulation is performed using a PUT endpoint.

Existing records can be altered, and the modified data directly replaces the original data.

There is no record of the previous version, meaning manipulation cannot be detected.

This clearly illustrates the vulnerability of centralized systems to insider corruption.

SLIDE 11 – SYSTEM B RESULTS (Jonathan)

For System B, every data submission requires explicit approval through the MetaMask wallet.

Each transaction generates a unique transaction hash, which is permanently recorded on the blockchain.

These records can be verified publicly using blockchain explorers such as Etherscan.

As a result, data integrity and accountability are fully preserved.

SLIDE 12 – CONCLUSION & RECOMMENDATION (Jonathan & William)

In conclusion, our experimental results show that blockchain-based systems significantly outperform centralized systems in preventing data manipulation.

Blockchain ensures transparency, immutability, and trust through cryptographic mechanisms.

Therefore, we recommend the gradual adoption of blockchain technology in e-government systems, particularly in areas requiring high transparency such as public fund management and procurement.

Thank you for your attention.